



## FP2 Chapter 10 Differentiation

**Lecture 10 Differentiation**

(Inverse Trigonometry, Hyperbolic Functions, Inverse Hyperbolic Functions, Implicit Differentiation, Parametric Differentiation, Maclaurin Series)

**1. Differentiation of Inverse Trigonometry Function**

- To differentiate  $y = \sin^{-1} x$ ,

Let  $\sin y = x$ ,

By implicit differentiation,

$$\cos y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}$$

- To differentiate  $y = \cos^{-1} x$ ,

Let  $\cos y = x$ ,

By implicit differentiation,

$$-\sin y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1 - \cos^2 y}} = -\frac{1}{\sqrt{1 - x^2}}$$

- To differentiate  $y = \tan^{-1} x$ , (shown in P3)

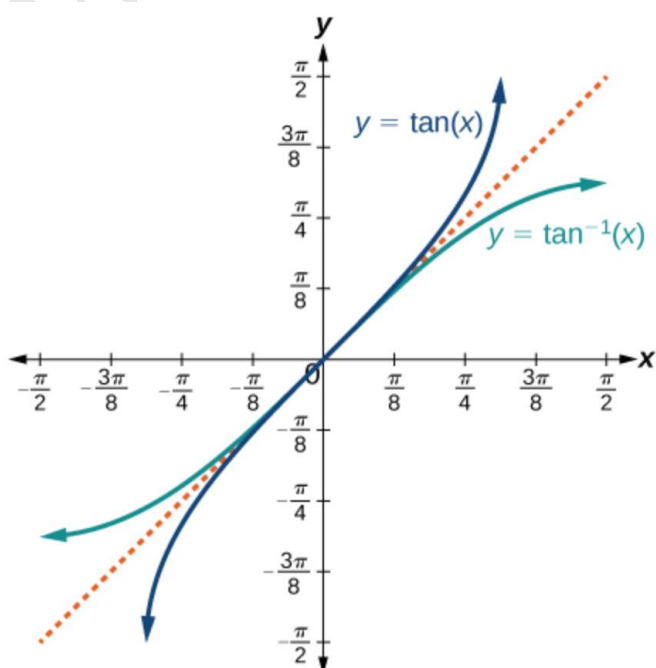
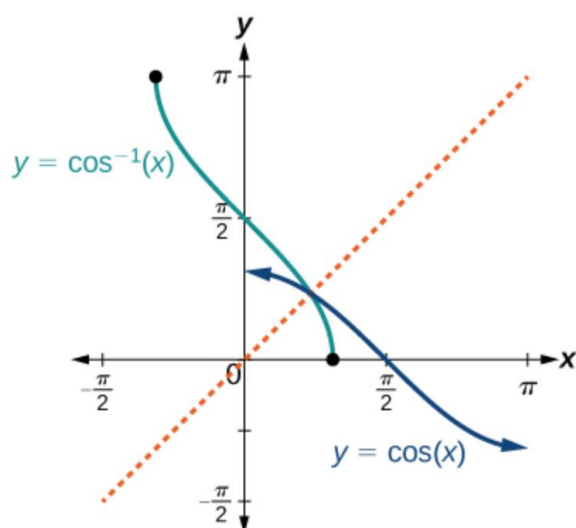
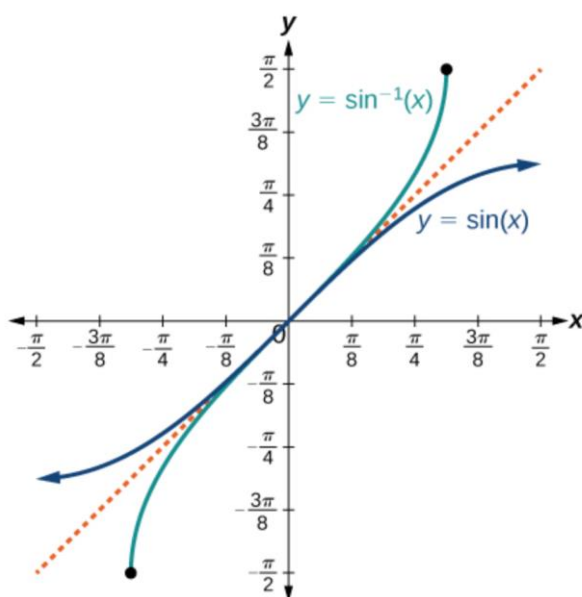
Let  $\tan y = x$ ,

By implicit differentiation,

$$\sec^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

- Remember that the trigonometry functions,  $\sin x$ ,  $\cos x$  and  $\tan x$  are not one-to-one function. Therefore they cannot have an inverse unless we restrict the domain.



- **Example 1: Differentiate the following functions.**

(i)  $y = \sin^{-1}\left(\frac{x}{3}\right)$

(ii)  $y = \cos^{-1}\sqrt{x}$

(iii)  $y = (\sin^{-1} x)^2$

(iv)  $y = \sin^{-1} x \cos^{-1} x$

(v)  $y = \sin^{-1}(x + 3)$

(vi)  $y = (\sin^{-1} 2x)^3$

- **Exercise A:**

1. Differentiate the following functions.

(i)  $y = \sin^{-1}(3x)$

(ii)  $y = x \sin^{-1} x$

(iii)  $y = \cos^{-1}(x^2)$

(iv)  $y = e^x \sin^{-1} x$

(v)  $y = \sin^{-1} \sqrt{2} x$

2. The function  $y = x - \sqrt{1 - x^2} \sin^{-1} x$ . Show that

$$\frac{dy}{dx} = \frac{x \sin^{-1} x}{\sqrt{1 - x^2}}.$$

**2. Differentiation of Hyperbolic Function**

- To differentiate  $y = \sinh x$ ,

$$y = \sinh x = \frac{1}{2}(e^x - e^{-x})$$

$$\frac{dy}{dx} = \frac{d}{dx}(\sinh x) = \frac{1}{2} \frac{d}{dx}(e^x - e^{-x}) = \frac{1}{2}(e^x + e^{-x}) = \cosh x$$

- To differentiate  $y = \cosh x$ ,

$$y = \cosh x = \frac{1}{2}(e^x + e^{-x})$$

$$\frac{dy}{dx} = \frac{d}{dx}(\cosh x) = \frac{1}{2} \frac{d}{dx}(e^x + e^{-x}) = \frac{1}{2}(e^x - e^{-x}) = \sinh x$$

- To differentiate  $y = \tanh x$ ,

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(\tanh x) = \frac{d}{dx} \left( \frac{\sinh x}{\cosh x} \right) = \frac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x} = \frac{1}{\cosh^2 x} \\ &= \operatorname{sech}^2 x \end{aligned}$$

- **Example 2: Differentiate the following functions.**

(i)  $y = \operatorname{sech} x$

(ii)  $y = \cosh 3x$

(iii)  $y = \tanh 2x$

(iv)  $y = \sinh 4x$

(v)  $y = \sinh^2 3x$

(vi)  $y = \sqrt{\cosh 5x}$

- **Exercise B: Differentiate the following functions.**

(i)  $y = x \sinh 2x$

(ii)  $y = \cosh 2x \tanh 2x$

(iii)  $y = \ln(\cosh x)$

(iv)  $y = e^{2x} \cosh 3x$

(v)  $y = \frac{1}{2} \sinh 4x + 4 \sinh 2x + 6x$ , show  $\frac{dy}{dx} = 16 \cosh^4 x$ .



### 3. Differentiation of Inverse Hyperbolic Function

- To differentiate  $y = \sinh^{-1} x$ ,

Let  $\sinh y = x$ ,

By implicit differentiation,

$$\cosh y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1 + \sinh^2 y}} = \frac{1}{\sqrt{1 + x^2}}$$

- Alternately, you can find the derivative from the logarithmic form.

$$\begin{aligned} \frac{d}{dx}(\sinh^{-1} x) &= \frac{d}{dx} \left[ \ln(x + \sqrt{x^2 + 1}) \right] = \frac{1}{x + \sqrt{x^2 + 1}} \left( 1 + \frac{x}{\sqrt{x^2 + 1}} \right) \\ &= \frac{1}{x + \sqrt{x^2 + 1}} \left( \frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}} \right) = \frac{1}{\sqrt{x^2 + 1}} \end{aligned}$$

- To differentiate  $y = \cosh^{-1} x$ ,

Let  $\cosh y = x$ ,

By implicit differentiation,

$$\sinh y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sinh y} = \frac{1}{\sqrt{\cosh^2 y - 1}} = \frac{1}{\sqrt{x^2 - 1}}$$

- To differentiate  $y = \tanh^{-1} x$ ,

Let  $\tanh y = x$ ,

By implicit differentiation,

$$\operatorname{sech}^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1 - \tanh^2 y} = \frac{1}{1 - x^2}$$



- **Example 3:**

Given that  $y = \sinh^{-1} x + \tanh^{-1} x$  ( $-1 < x < 1$ ), show that  $\frac{dy}{dx} = 0$  for only one value of  $x$ .

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- **Example 4:**

It is given that  $y = \operatorname{cosech}^{-1}x$ . Show that for  $x > 0$ ,  $\frac{dy}{dx} = -\frac{1}{x\sqrt{x^2+1}}$  and write down an expression for  $\frac{dy}{dx}$  in terms of  $x$  for  $x < 0$ .

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- **Example 5:**

Prove that  $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$ . Given that  $y = (\coth^{-1} x)^2$ , show that

$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = \frac{k}{1-x^2}$ , where  $k$  is a constant to be determined.

Sunway A Levels

- **Example 6:**

It is given that  $y = \frac{1}{2} \ln(\coth x)$ ,  $x > 0$ . Show that  $\frac{dy}{dx} = -\operatorname{cosech}(2x)$ .

- **Example 7:**

Given that  $y = \sinh^{-1}(\tanh x)$ , show that  $\frac{dy}{dx} = \frac{\operatorname{sech}^2 x}{\sqrt{1+\tanh^2 x}}$ .

- **Exercise C: Differentiate the following functions.**

(i)  $y = \sinh^{-1} 5x$

(ii)  $y = \cosh^{-1} 3x$

(iii)  $y = \sinh^{-1} \sqrt{2}x$

(iv)  $y = \cosh^{-1} \frac{3}{4}x$

(v)  $y = \sinh^{-1} x^2$

(vi)  $y = \operatorname{sech}^{-1} x$

4. To obtain an expression for  $\frac{d^2y}{dx^2}$  in the case where the relation between  $y$  and  $x$  is defined implicitly.

- **Example 8:**

Find the value of  $\frac{d^2y}{dx^2}$  at the point (1, 2) on the curve  $2x^3 + 2y^3 - 9xy = 0$ .

- **Example 9:**

Show that  $\frac{dy}{dx} = 1$  at the point  $A(1,1)$  on the curve  $y + x \ln y - x^2 = 0$ . Find the value of  $\frac{d^2y}{dx^2}$  at  $A$ .

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- **Example 10:**

Show that  $\frac{dy}{dx} = -\frac{1}{2}$  at the point  $A\left(1, \frac{1}{2}\right)$  on the curve  $x + 2y + \tan^{-1}(2xy) = 2 + \frac{\pi}{4}$ . Find the value of  $\frac{d^2y}{dx^2}$  at  $A$ .

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- **Example 11:**

Given that the equation of a curve is  $y = x + \tanh^{-1}(x + y)$ , show that, at all points of the curve except the origin,

(i)  $\frac{dy}{dx} = 1 - \frac{2}{(x+y)^2}$

(ii)  $\frac{d^2y}{dx^2} = 8 \cosh^3(x - y) \operatorname{cosech}^5(x - y).$

- **Example 12 (PYQ O/N 2016/11/12/13/8):**

A curve  $C$  has equation  $x^2 + 4xy - y^2 + 20 = 0$ . Show that, at stationary points on  $C$ ,  $x = -2y$ . [3]

Find the coordinates of the stationary points on  $C$ , and determine their nature by considering the value of  $\frac{d^2y}{dx^2}$  at the stationary points. [8]

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- **Example 13 (PYQ O/N 2006/1/10):**

The curve  $C$  has equation

$$y = x^2 + \lambda \sin(x + y),$$

where  $\lambda$  is a constant, and passes through the point  $A\left(\frac{1}{4}\pi, \frac{1}{4}\pi\right)$ . Show that  $C$  has no tangent which is parallel to the  $y$ -axis. [5]

Show that, at  $A$ ,

$$\frac{d^2y}{dx^2} = 2 - \frac{1}{64}\pi(4 - \pi)(\pi + 2)^2. \quad [5]$$

- **Exercise D:**

1. The curve  $C$  has equation  $xy - \ln(x + y^2 - 10) - 6 = 0$ .
  - (i) Show that, at any point of  $C$ ,  $\left(x \frac{dy}{dx} + y\right)(x + y^2 - 10) = 1 + 2y \frac{dy}{dx}$ .
  - (ii) Find the value of  $\frac{d^2y}{dx^2}$  at the point  $(2, 3)$ .
2. A curve has equation  $(x^2 + y^2)^2 = 16(x^2 - y^2)$ .
  - (i) Show that  $(y^3 + 8y + x^2y) \frac{dy}{dx} = 8x - xy^2 - x^3$ .
  - (ii) Find the coordinates of the stationary points of the curve.
  - (iii) Find the coordinates of the points where the tangent is vertical.
  - (iv) By finding the second derivative, classify the stationary points.

3. A curve has equation  $\sin 2x + \sin y = 1$ .

(i) Find the gradient at the point where  $x = \frac{\pi}{3}$ .

(ii) Find the value of  $\frac{d^2y}{dx^2}$  at this point.

4. A curve has equation  $xe^{\sin y} = 3$ .

(i) Find an expression for  $\frac{d^2y}{dx^2}$ .

(ii) Find the value of  $\frac{d^2y}{dx^2}$  when  $x = 4$ .



5. To obtain an expression for  $\frac{d^2y}{dx^2}$  in the case where the relation between  $y$  and  $x$  is defined parametrically.

- If a curve is defined by the parametric equations  $x = f(t), y = g(t)$ , then

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left( \frac{dy}{dx} \right), \text{ where } \frac{dy}{dx} \text{ is a function of } t$$

$$= \frac{d}{dt} \left( \frac{dy}{dx} \right) \frac{dt}{dx}$$

- **Example 14:**

A curve is defined parametrically by  $x = t^2 - 2 \ln t, y = 4(t - 1), t \geq 1$ . Find  $\frac{d^2y}{dx^2}$  in terms of  $t$ .

- **Example 15:**

A curve is defined parametrically by  $x = \cos 2t$ ,  $y = 2t + \sin 2t$ . Show that  $\frac{dy}{dx} = -\cot t$  and hence determine the value of  $\frac{d^2y}{dx^2}$  in terms of  $t$ .

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- **Example 16:**

Given that  $x = a(t - \sin t)$  and  $y = a(1 - \cos t)$  where  $a$  is a positive constant, find  $\frac{dy}{dx}$  in terms of  $t$ , and show that  $\frac{d^2y}{dx^2} = -\frac{1}{4a} \operatorname{cosec}^4\left(\frac{1}{2}t\right)$ .

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- **Example 17:**

The curve  $C$  is defined parametrically for all real values of  $t$  by the equations  
 $x = t + \cosh t, y = \sinh t$ .

(a) Show that  $\frac{d^2y}{dx^2} = \frac{\sinh(t)-1}{(\sinh(t)+1)^3}$ .

(b) Hence find the coordinates of the point on  $C$  at which  $\frac{d^2y}{dx^2} = 0$ .

- **Example 18:**

The curve  $C$  has parametric equations  $x = \sin t, y = \sin 2t$  for  $0 \leq t \leq \pi$ .

Find  $\frac{d^2y}{dx^2}$  in terms of  $t$ .

Hence, or otherwise, find the coordinates of the stationary points on  $C$  and determine their nature.

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- **Example 19 (PYQ O/N 2009/1/4):**

It is given that

$$x = t + \sin t, \quad y = t^2 + 2 \cos t,$$

where  $-\pi < t < \pi$ . Find  $\frac{dy}{dx}$  in terms of  $t$ . [2]

Show that

$$\frac{d^2y}{dx^2} = \frac{2t \sin t}{(1 + \cos t)^3}. \quad [4]$$

Show that  $\frac{dy}{dx}$  increases with  $x$  over the given interval of  $t$ . [2]

- **Example 20 (PYQ O/N 2013/13/4):**

A curve has parametric equations

$$x = 2\theta - \sin 2\theta, \quad y = 1 - \cos 2\theta, \quad \text{for } -3\pi \leq \theta \leq 3\pi.$$

Show that

$$\frac{dy}{dx} = \cot \theta,$$

except for certain values of  $\theta$ , which should be stated.

[4]

Find the value of  $\frac{d^2y}{dx^2}$  when  $\theta = \frac{1}{4}\pi$ .

[3]

- **Exercise E:**

1. A curve is defined by the parametric equations  $x = t^2 - t + 1, y = t^3 - 3t - 1$ .
  - (a) Find the coordinates of the stationary points of the curve.
  - (b) Identify the nature of these stationary points.
  - (c) Show that the curve passes through one of stationary points twice.
2. A curve is defined by the parametric equations  $x = t + \sin 2t, y = \cos 2t$ .
  - (a) Show that  $\frac{dy}{dx} = \frac{-2 \sin 2t}{1 + 2 \cos 2t}$ .
  - (b) Show that  $\frac{d^2y}{dx^2} = \frac{-4(y+2)}{(1+2y)^3}$ .

3. A curve is defined by the parametric equations  $x = 1 + 2 \cos t$ ,  $y = \cos t + \sin t$ .
- (a) Show that  $\frac{dy}{dx} = \frac{\tan(t)-1}{2 \tan t}$ .
- (b) Show that  $\frac{d^2y}{dx^2} = \frac{-\sec^2 t}{4 \tan^3 t}$ .
- (c) Hence find and classify all the stationary points of the curve.
4. The curve  $C$  is defined parametrically by  $x = \frac{3t}{1+t^2}$ ,  $y = \frac{3t^2}{1+t^3}$ , where  $t$  is a real parameter and  $t \neq -1$ . Given that  $t \neq 2^{-\frac{1}{3}}$ , show that  $\frac{dy}{dx} = \frac{2t-t^4}{1-2t^3}$ .

Show also that there is no point of  $C$  at which  $\frac{d^2y}{dx^2} = 0$ .



## 6. Maclaurin's Series

- If  $f(x)$  can be expanded in a convergent series of positive integral powers of  $x$ , then

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

MF19

- **Proof:** Let  $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$

Differentiating repeatedly with respect to  $x$  gives

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots$$

$$f''(x) = 2a_2 + 6a_3x + 12a_4x^2 + \dots$$

$$f'''(x) = 6a_3 + 24a_4x + \dots$$

$$a_0 = f(0); a_1 = f'(0); a_2 = \frac{1}{2} f''(0) = \frac{1}{2!} f''(0); a_3 = \frac{1}{6} f'''(0) = \frac{1}{3!} f'''(0); \dots$$

$$\therefore f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

- **Example 21:** Find Maclaurin series for  $e^x$ .

$$\text{let } f(x) = e^x \quad \rightarrow f(0) = e^0 = 1$$

$$f'(x) = e^x \quad \rightarrow f'(0) = e^0 = 1$$

$$f''(x) = e^x \quad \rightarrow f''(0) = e^0 = 1$$

$$\text{In general, } f^{(n)}(x) = e^x, n \geq 1 \rightarrow f^{(n)}(0) = 1$$

$$e^x = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots$$



- **Example 22:** Find Maclaurin series for  $\ln(1+x)$ .

$$\text{Let } f(x) = \ln(1+x) \rightarrow f(0) = 0$$

$$f'(x) = \frac{1}{1+x} \rightarrow f'(0) = 1$$

$$f''(x) = -\frac{1}{(1+x)^2} \rightarrow f''(0) = -1$$

$$f'''(x) = \frac{2}{(1+x)^3} \rightarrow f'''(0) = 2$$

$$\begin{aligned} \ln(1+x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots \\ &= 0 + x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots \end{aligned}$$

- **Example 23:** Find Maclaurin series for  $\sin x$ .

$$\text{Let } f(x) = \sin x \rightarrow f(0) = 0$$

$$f'(x) = \cos x \rightarrow f'(0) = 1$$

$$f''(x) = -\sin x \rightarrow f''(0) = 0$$

$$f'''(x) = -\cos x \rightarrow f'''(0) = -1$$

$$\begin{aligned} \sin x &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \end{aligned}$$





- **Example 24:** Find Maclaurin series for  $\cos x$ .

rule  $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

### Standard Expansion (in MF19)

Maclaurin's series:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^r}{r!}f^{(r)}(0) + \dots$$

$$e^x = \exp(x) = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^r}{r!} + \dots$$

(all  $x$ )

$\ln x$  ?

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{r+1} \frac{x^r}{r} + \dots$$

$(-1 < x \leq 1)$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^r \frac{x^{2r+1}}{(2r+1)!} + \dots$$

(all  $x$ )

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^r \frac{x^{2r}}{(2r)!} + \dots$$

(all  $x$ )

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots + (-1)^r \frac{x^{2r+1}}{2r+1} + \dots$$

$(-1 \leq x \leq 1)$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{x^{2r+1}}{(2r+1)!} + \dots$$

(all  $x$ )

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{x^{2r}}{(2r)!} + \dots$$

(all  $x$ )

$$\tanh^{-1} x = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots + \frac{x^{2r+1}}{2r+1} + \dots$$

$(-1 < x < 1)$



- \* - If  $f(x)$  is undefined at  $x = 0$ , i.e.  $f(0) = \infty$  or  $-\infty$ , then  $f(x)$  cannot be expressed as a Maclaurin series.
- Notice that  $\frac{1}{x}$ ,  $\ln x$  and  $\cot x$  cannot be expressed by Maclaurin series because they are undefined at  $x = 0$ .
- There are two ways to find Maclaurin series of certain function.
  - (i) Repeat the process step by step as in the previous examples.
  - (ii) With the help of MF19.

**Example 25:**

$$f(x) = \tan(e^x - 1)$$

If  $y = \tan(e^x - 1)$ , prove that  $\frac{d^2y}{dx^2} = \frac{dy}{dx}(1 + 2e^xy)$ . Obtain the Maclaurin expansion of  $y$  as far as the term in  $x^3$ .

$$y = \tan(e^x - 1)$$

$$\frac{dy}{dx} = \sec^2(e^x - 1) \cdot e^x$$

$$= e^x \sec^2(e^x - 1)$$

$$\frac{d^2y}{dx^2} = e^x \sec^2(e^x - 1) + 2 \sec(e^x - 1) \cdot \sec(e^x - 1) \tan(e^x - 1) \cdot e^x \cdot e^x$$

$$= e^x \sec^2(e^x - 1) + 2e^{2x} \sec^2(e^x - 1) \tan(e^x - 1)$$

$$= e^x \sec^2(e^x - 1) [1 + 2e^x \tan(e^x - 1)]$$

$$= \frac{dy}{dx} [1 + 2e^xy]$$

$$f''(x) = f'(x) [1 + 2e^x f(x)]$$

when  $x=0$ ,  $f(0) = \tan(e^0 - 1) = 0$

$$f'(0) = e^0 \sec^2(e^0 - 1) = 1$$

$$f''(0) = (1) [1 + 2e^0(0)] = 1$$

$$f'''(x) = f''(x) [1 + 2e^x f(x)] + [2e^x f(x) + f'(x)e^x] f'(x)$$

$$f'''(0) = (1) [1 + 2e^0(0)] + [2e^0(0) + (1)e^0] (1) = 1 + 2 = 3$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 = 0 + x + \frac{1}{2}x^2 + \frac{3}{6}x^3 = x + \frac{1}{2}x^2 + \frac{1}{2}x^3$$



- **Example 26:**

If  $y = \ln(\sec x + \tan x)$ , prove that  $e^y \frac{dy}{dx} = \left(\frac{dy}{dx}\right)^2 + \frac{d^2y}{dx^2}$ . Expand  $y$  in ascending powers of  $x$  as far as the term in  $x^3$ .

$$y = \ln(\sec x + \tan x)$$

$$e^y = \sec x + \tan x$$

$$\begin{aligned} e^y \frac{dy}{dx} &= \sec x \tan x + \sec^2 x \\ &= \sec x (\tan x + \sec x) \\ &= e^y \sec x \end{aligned}$$

$$\frac{dy}{dx} = \sec x$$

$$\frac{d^2y}{dx^2} = \sec x \tan x$$

$$e^y \frac{dy}{dx} = \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2$$

$$e^y f'(x) = f''(x) + [f'(x)]^2$$

$$\ln(\sec x + \tan x) = x + \frac{x^3}{6}$$

- **Example 27:**

It is given that  $f(x) = \tanh^{-1}\left(\frac{1-x}{3+x}\right)$  for  $x > -1$ .

(a) Show that  $f''(x) = \frac{1}{2(x+1)^2}$ .

(b) Hence find the Maclaurin series for  $f(x)$  up to and including the term in  $x^2$ .

Sunway A Levels

- **Example 28:**

If  $y = (\sin^{-1} x)^2$  and  $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$ , Show that  $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2$ . Obtain the expansion of  $y$  as far as the term in  $x^4$ .

Sunway A Levels

- **Example 29:**

It is given that  $y = \sinh(2 \sinh^{-1} x)$ , show that

(a)  $(1 + x^2) \left( \frac{dy}{dx} \right)^2 - 4y^2 = 4$

(b)  $(1 + x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - 4y = 0.$

Hence expand  $y$  in ascending powers of  $x$  as far as the term in  $x^5$ .

- **Exercise F:**

1. Given that  $y = e^{\sin^{-1} x}$ , prove that  $(1 - x^2) \frac{d^2 y}{dx^2} = y + x \frac{dy}{dx}$ . Hence expand  $y$  in ascending powers of  $x$  up to the term in  $x^3$ .
2. Given that  $y = \cos^{-1} x$ , prove that  $(x^2 - 1) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 0$ . Hence find the first three non-zero terms in series expansion for  $\cos^{-1} x$ .

3. Given that  $y = e^{\sqrt{1+x}}$ , prove that  $4(1+x)\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = y$ . Hence expand  $y$  in ascending powers of  $x$  up to the term in  $x^4$ .
4. Given that  $y = e^{-3x} \sin 2x$ , show that  $\frac{dy}{dx} = 2e^{-3x} \cos 2x - 3y$  and  $\frac{d^2y}{dx^2} = -13y - 6\frac{dy}{dx}$ . Hence, find the Maclaurin series for  $y$  up to the term in  $x^4$ .

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5. Given that  $y = \tan\left(2x + \frac{4}{3}\pi\right)$ , show that  $\frac{dy}{dx} = 2(1 + y^2)$  and  $\frac{d^3y}{dx^3} = 4\left[\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}\right]$ . Hence, find the Maclaurin series for  $y$  up to the term in  $x^3$ .
6. Given that  $y = \frac{1}{1+\sin 2x}$ , find the values of  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$  when  $x = 0$ . Hence find the first three non-zero terms in series expansion for  $y$ .



7. Find the Maclaurin series expansion for  $y$  up to and including the term  $x^2$  given that  $\frac{dy}{dx} = y^3 + x^8$  and  $y = 1$  when  $x = 0$ .

- 8.(a) Find the first three non-zero terms of the Maclaurin series for

$$\frac{e^{-2x} \sin x}{\ln(1+x)}.$$

Then, state the range of values of  $x$  for which the series is valid.

- (b) Using the result that  $\tan^{-1} x = \int_0^x \frac{1}{1+t^2} dt$ , show that  $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ , where  $|x| < 1$ . Hence

$$\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{\sin 2x}.$$

9. Write down and simplify the first three terms in the Maclaurin's expansion of  $e^{2x}$ .

Hence, show that the Maclaurin's series for  $\ln(e^{2x} + e^{-2x})$  begins with  $\ln a + bx^2$ , where  $a$  and  $b$  are constants to be found.

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